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The Editors

Superconductor Science and Technology

Submission of a paper: “Damage–repair kinetics and phase stiffness in a site-diluted XY model: implications for irradiated superconductors”

Dear Editors,

I submit the enclosed paper for consideration in Superconductor Science and Technology. It asks what a minimal model can say about the competition between displacement damage and recovery in irradiated superconductors, the subject of recent work published in this journal on REBCO conductors for fusion magnets [Prokopec et al 2015, Fischer et al 2018, Unterrainer et al 2022, Adams et al 2023]. In those studies the transition temperature falls linearly with displacement damage, part of it is recovered by annealing, and the degradation is attributed to a loss of superfluid density. The paper takes the phase stiffness, the counterpart of the superfluid density, as its observable.

Main results

- A mean-field reduction of damage–repair kinetics, re-derived from the implemented update for the damaged fraction, predicts bistability. On the lattice, with up to 10 seeds per condition and sizes to $L = 64$, no hysteresis and no dependence on the initial state is resolved (largest difference 0.0027 in the coherence, none significant after Holm correction; quasi-static ramps close), because the closure of the reduction fails: the multiplier of the damage rate that the lattice applies is a smooth function of the damaged fraction. The statement is made for the ranges tested.
- Without a field, and with repair that preserves the phase symmetry, repair restores spontaneous phase coherence and a stiffness, measured as the twist response of the non-equilibrium steady state with the winding number recorded, above a threshold ratio of repair to damage, in the thermodynamic limit at $\rho = 7.00\text{--}7.75$ (repair that leaves the phase unchanged) and $5.50\text{--}6.00$ (repair to the phase of the neighbors), against about 2.5 without state-dependent damage. The equilibrium helicity-modulus formula underestimates this stiffness by up to about 30% and is negative where the stiffness vanishes.
- The repair-to-damage ratio controls the state only when damage is slow compared with phase relaxation: three repair rules that differ by 0.12 in the coherence at finite rates agree within 0.012 as the damage rate falls, and the equilibrium formula converges to the twist response.
- Static random dilution of the lattice loses stiffness at a removed fraction of 0.25 ± 0.01 (sizes to 96, finite-size extrapolation and its sensitivity reported); compact damage clusters tolerate considerably more. An illustrative mapping to displacement damage gives about 10 model sites per dpa, and an illustrative annealing analysis shows that a recovery growing linearly over $150\text{--}400\text{ }^{\circ}\text{C}$ needs an activation spectrum of order 1 eV or more in width.

The model is phenomenological and is not calibrated to any conductor; it has no pinning, so it says nothing about the critical current or its initial rise with fluence, and I have said so in the paper. The experimental values quoted are taken from the cited papers.

Data, code and disclosures

All code, raw data and analysis scripts are public at <https://github.com/sandlerleon/damage-repair-xy-stiffness>, archived at <https://doi.org/10.5281/zenodo.23138195>; the manuscript is available as a preprint at <https://doi.org/10.5281/zenodo.23138197>. The simulation engine reproduces the update rules of the open-source MyUncle framework bit for bit. A separate manuscript on the same order-dependent-degradation mechanism in a rotor lattice without superconducting context (preprint <https://doi.org/10.5281/zenodo.21210708>) is intended for a different journal; the present paper does not depend on its conclusions. This manuscript has not been published and is not under consideration by any other journal. I declare no competing interests and no funding, and the use of AI-assisted tools is declared in the manuscript.

Thank you for considering the paper.

Yours sincerely,

Leon Sandler